Milaan

Closing the books before noon

The Singleton · Hemanth Chilakala

ExaThon India 2026 · Track 1: Money, Risk & Regulation

An accountable close, powered by Exasol Personal.

Meet the reconciliation analyst

> A reconciliation analyst at a mid-sized Indian NBFC can't close the day's books on time because UPI, NACH and payment-gateway settlement files arrive in different formats with mismatched references, so they chase breaks by hand in Excel. If this worked, they'd close the books before noon and spend their time only on the exceptions that actually matter.

Milaan reconciles synthetic receipts for the fictional Sahyog Finance Ltd, using Exasol Personal for normalisation, matching, scoring and audit records.

The person: a finance analyst at a mid-sized Indian NBFC.
The desired outcome: investigate meaningful exceptions instead of every receipt.

One credit can represent 37 payments

37 PayFast payments

gross ₹6,07,085.27

− fees ₹4,856.70 − GST ₹874.21

One bank credit: ₹6,01,354.36

Four channels · different formats · T+1/T+2 timing · truncated references · NACH returns

Measured synthetic showcase; 36 receipts are present and one is missing.

The cost of an unresolved close

Money in suspense delays explanation of the cash balance.

Misstated customer dues can prompt a call to someone who already paid.

Unclear evidence adds effort to accounting review and audit.

Illustrative workflow consequences. No industry cost or productivity statistic is claimed.

Everything heavy runs inside Exasol

Four source CSVs → Bulk IMPORT → SQL normalisation

↓

7 matching passes + UDFs → Ranked exceptions

↓

Streamlit analyst review → Exasol decision audit

Ground truth enters only the SQL scoring path. Python orchestrates; the database reconciles.

Seven passes, measured at 10M

Pass	Receipt links	Time
EXACT	3,164,644	13.16s
PG_BATCH	35,303	11.42s
FEE_AWARE	1,079,439	8.88s
TOLERANT	43,689	2.75s
FUZZY	16,025	2.26s
SPLIT_MERGE	44,100	16.95s
CLASSIFY	0	22.27s

Passes use remaining keys only. One-to-many and many-to-one relationships share a single group ID.

Why Exasol Personal

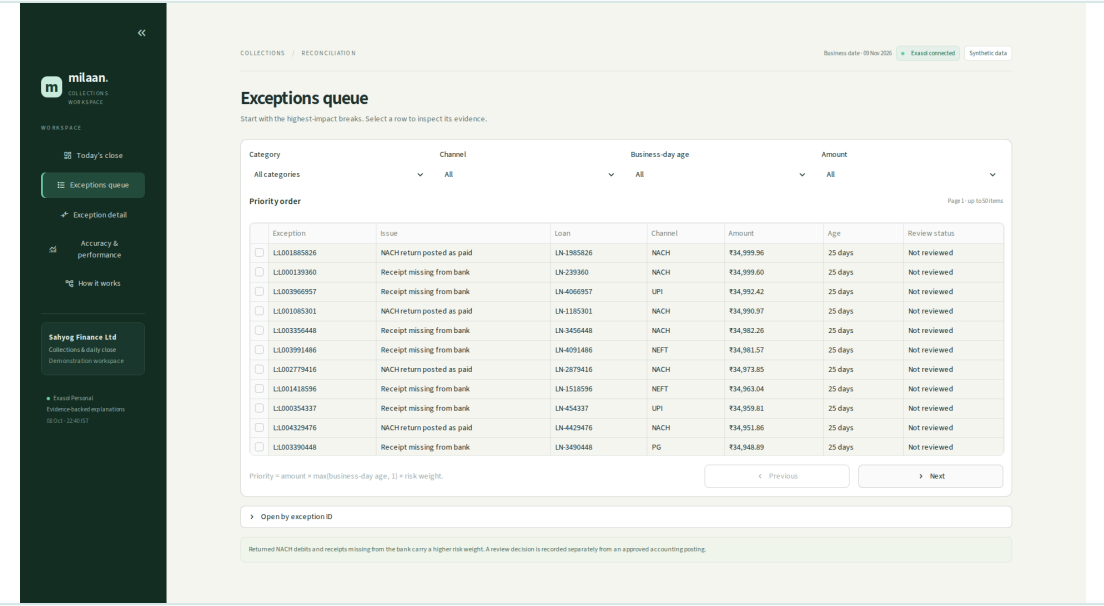
114.49 seconds

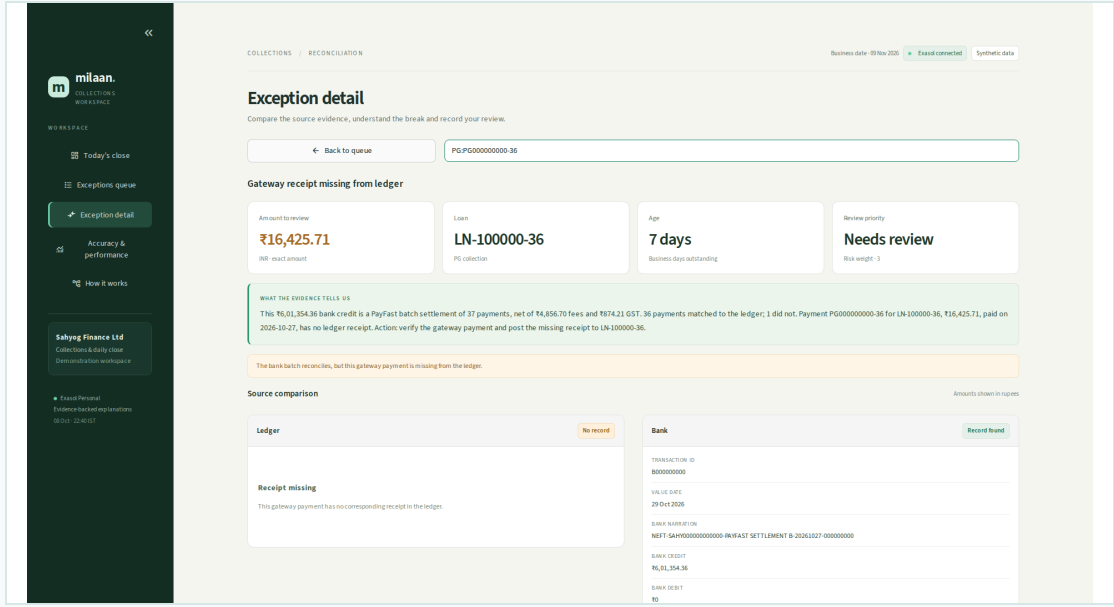
10,000,000 source records across four files.

SQL joins, aggregation and windows · bulk IMPORT · Python scalar and SET UDFs

Measured on 4 cores / 15.62 GiB RAM.
Timing includes results, conflict validation and SQL scoring; explanations and profiling flush are separate. Live-recorded run when available; timings vary between runs.

Review the money that needs attention





Rank by amount × business-day age × risk. Inspect source evidence. Accept a recommendation, reject it or assign an owner.

Accuracy against known answers

100.000%
precision

99.959%
recall

98.502%
source records grouped

0 false links · 1,779 missed expected links · 105,343 exceptions

Conflict validation rejects records assigned to multiple groups. These seeded results do not establish real-world accuracy; corruption patterns and distinct loans are deliberately constrained.

Trust comes from evidence

Deterministic explanations: amounts, dates and rules come from stored evidence.

Human review: decisions are recorded, not silently posted to a real ledger.

Run-linked audit: analyst, timestamp, assignment and note stay in Exasol.

No external inference API. PII masking is tested for a future optional rewrite; that integration is not shipped.

A practical path to operational impact

First: validate with an NBFC partner using independently labelled, anonymised cases.

Next: live SFTP/API feeds, real business calendars, review roles and approved ledger write-back.

Then: measure larger datasets and distributed coordination.

The same pattern may help banks, insurers, fintechs and gateway-based merchants. Applicability is a hypothesis to test.

Close the books with confidence

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Repository destination:
https://github.com/Hemanth-Chilakala/milaan

Demo: `docs/video/demo.mp4`

MIT · synthetic data only · Exasol Personal

Thank you.